

Temporomandibular Joint Disorder with Evc2 mutant mice, XROMM analysis on abnormal jaw movement and pain

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Introduction

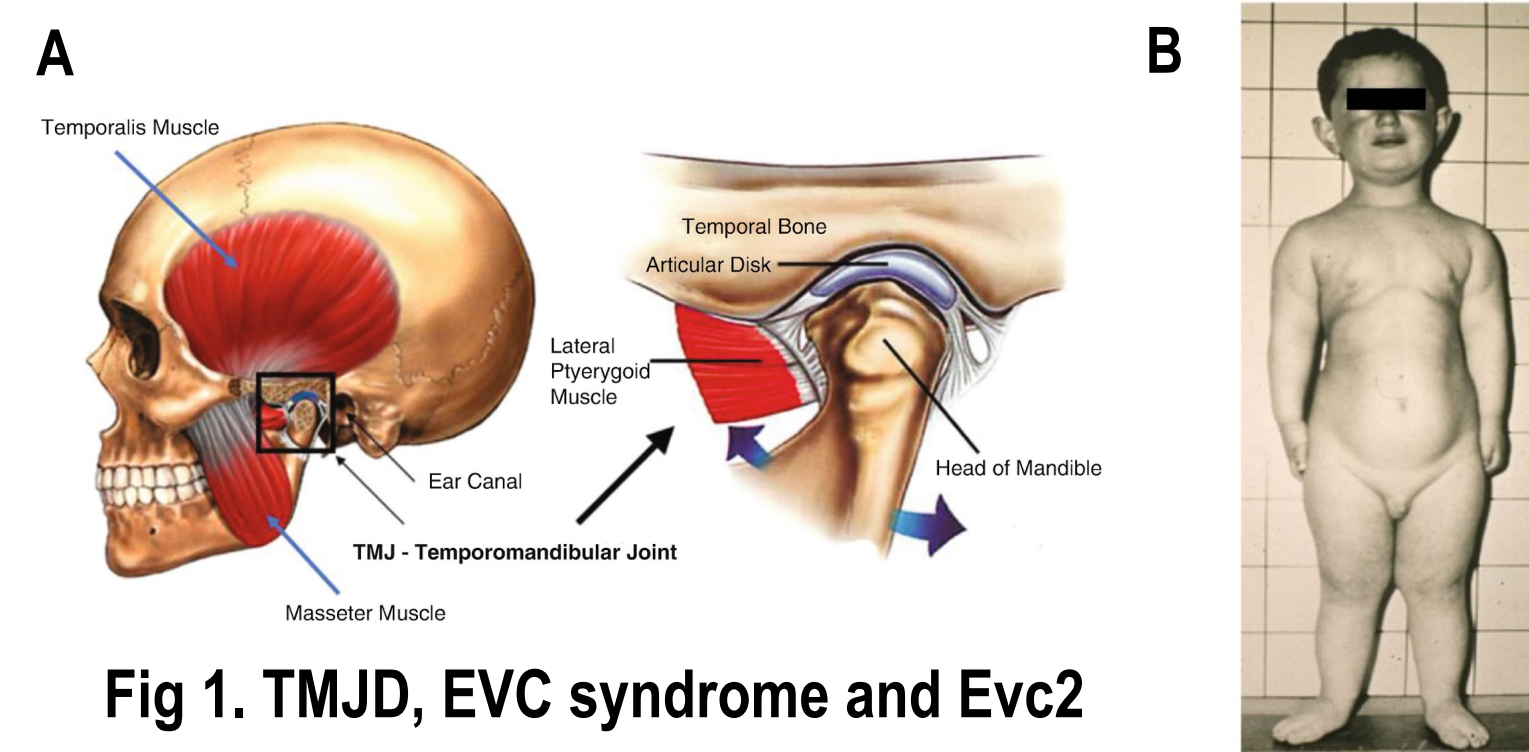


Fig 1. TMJD, EVC syndrome and Evc2

A. Temporomandibular joint disorders (TMD) are a set of disorders including painful conditions in temporomandibular joints (TMJ), muscles of mastication, and surrounding tissues. Since the etiology and pathophysiology of this disorder are elusive, it is crucial to testify if abnormal TMJ shape is a potential reason leading to muscle pain in TMD. The estimated prevalence in the US is 4.6%. Up to today, the actual etiology of TMD still remains to be fully elusive. This is, in part, due to lack of animal model for studies in understanding the etiology of TMD.

B. Ellis-van Creveld (EVC) syndrome is a chondrodysplasia. Affected individuals have unportioned shortened limbs compared to unaffected. To dissect the relation between TMJ shape, abnormal jaw movement, and pain, we generate Evc2 mutant mice to know the correlation between TMD symptoms and Evc2 mutation.

Objectives

Our question is if Evc2 mutant mice show symptoms that TMD patients would show, such as muscle pain, decreased displacement, decreased biting force, and difficulty when opening their mouth along with abnormally shaped TMJ.

Method



Fig 2. Process of Tracking Mandible and Temporal with X-ray reconstruction of moving morphology (XROMM)

To assess the jaw movement and digitalize locations and movements, we used a detection method called X-ray reconstruction of moving morphology, XROMM, and data were analyzed by a program called XMA Lab. XROMM is a functional tool for tracking small metal markers surgically implanted to the temporal (the sides of the skull) and mandible (lower jaw) by naming them 'top' and 'bottom' respectively (left).

We measured the bottom distance to evaluate levels of openness of the mouth, frequency of chew, and consistency of chewing cycles, which are indications of symptoms of TMD. We compare the jaw movement of the Evc2 mutant mice with control mice.

Result

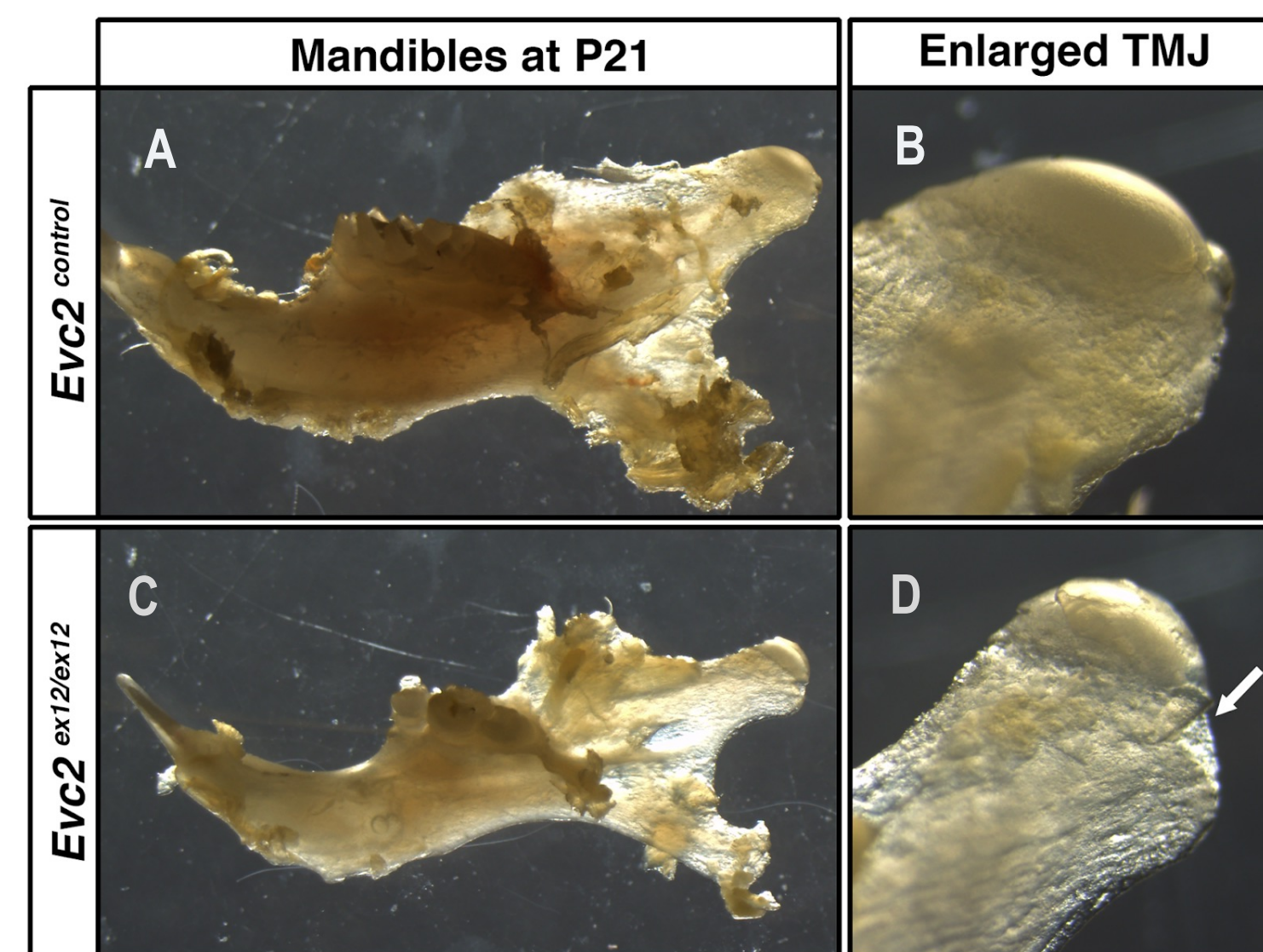


Fig 3. Evc2 loss-of-function leads to abnormally developed TMJ

Evc2 global knockout mutant mice have abnormally developed condylar cartilages. Mandibles of controls (A), Evc2 mutants (C) and enlarged regions (B and D) for condylar cartilages were shown. White arrow indicated the condylar bone with no cartilage covering.

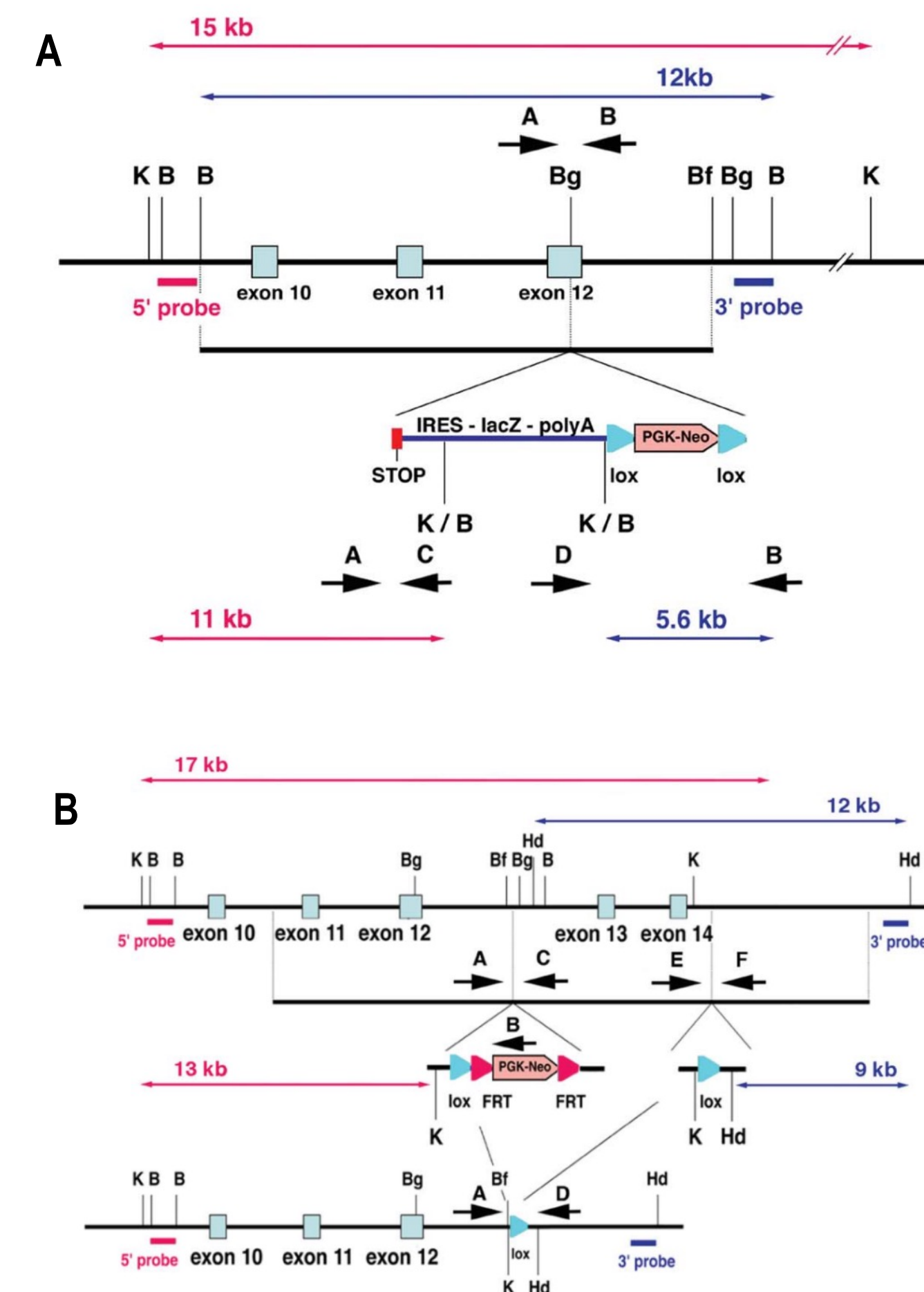


Fig 4. Generation of Evc2 mutant mouse lines

Evc2 mutant mice were generated by introducing a premature stop codon into exon12 of mouse Evc2, to mimic a nonsense mutation identified in patient. The IRES-lacZ cassette was also introduced downstream of the STOP codon for monitoring endogenous Evc2 expressions. A is a process of generating global knockout (loss-of-function) Evc2 mutant mice. B is a process of generating of a conditional knockout (loss-of-function) model.

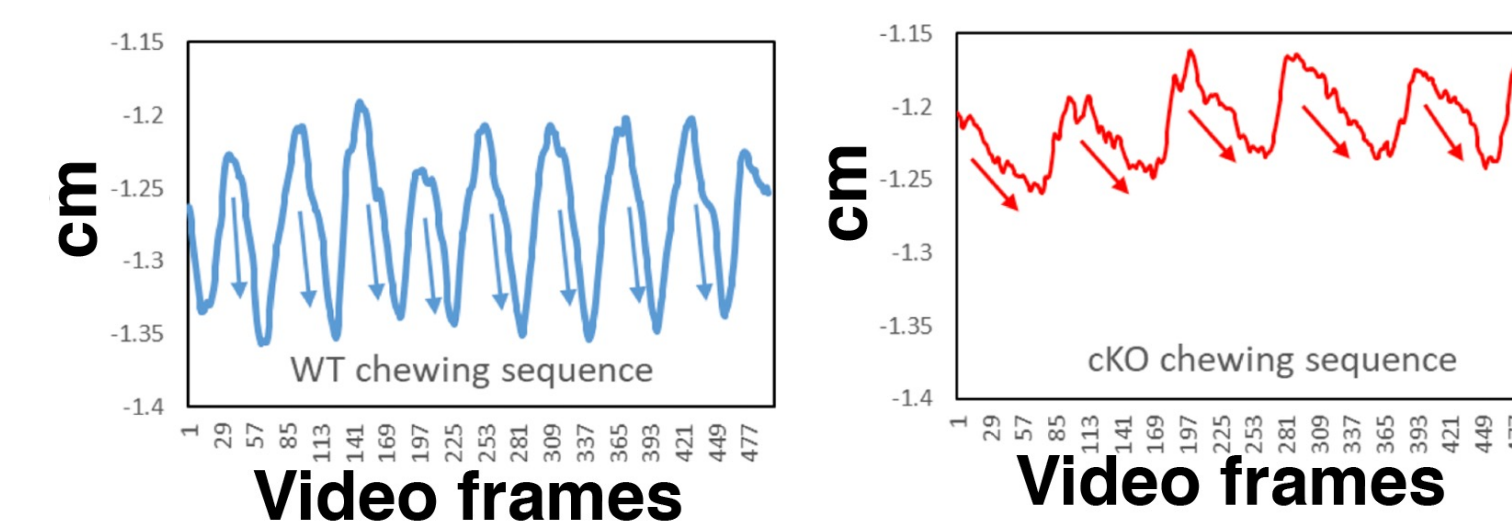
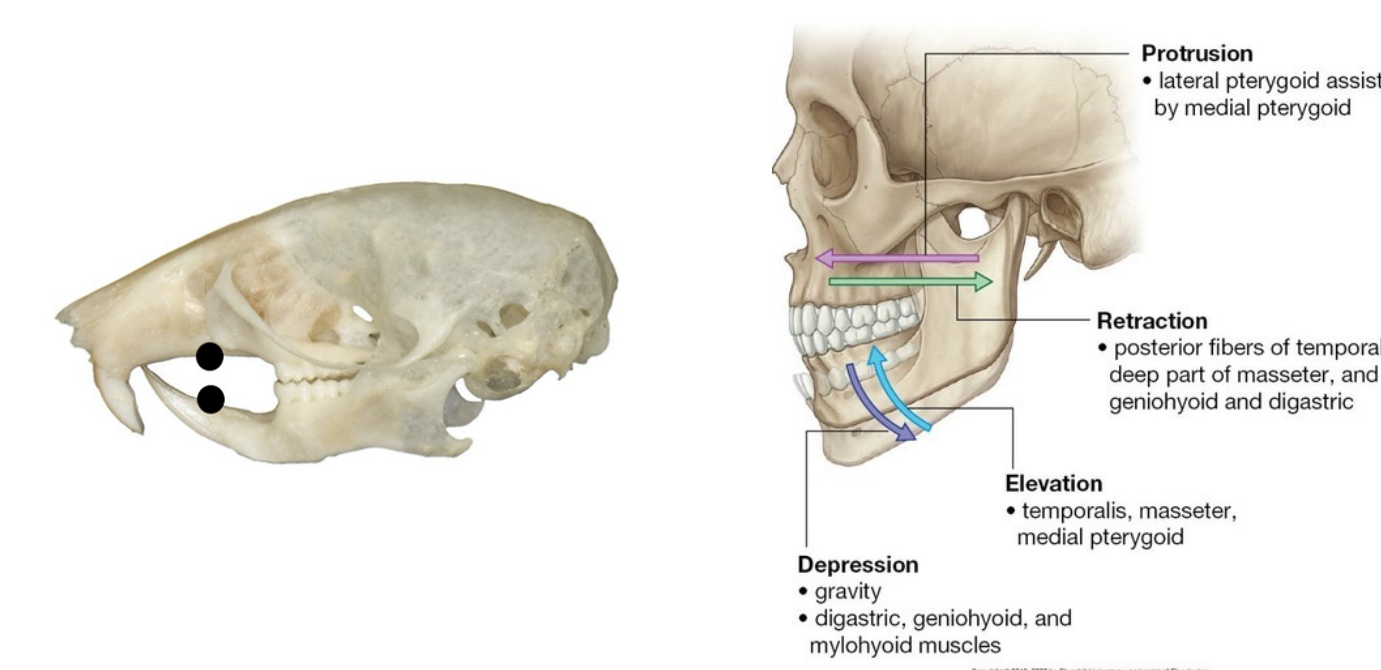
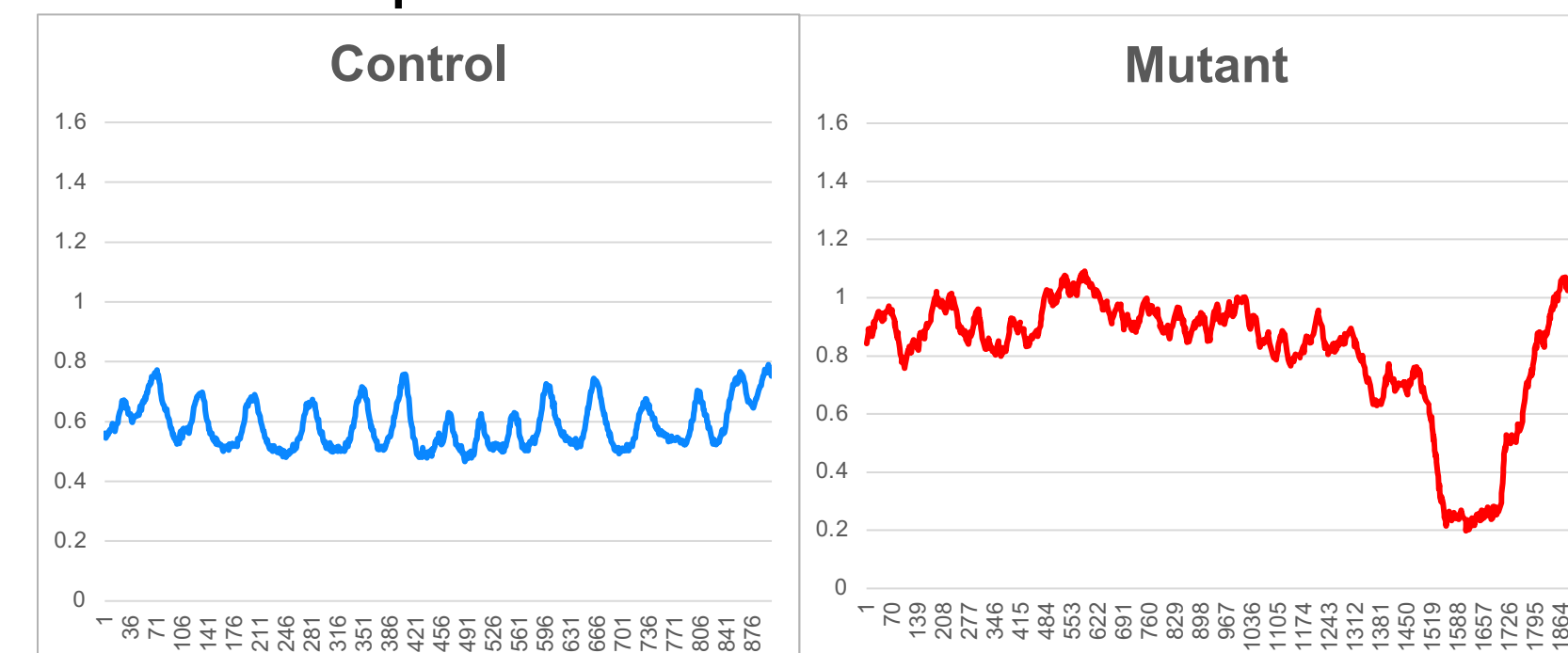


Fig 5. XROMM and abnormal jaw movement of Evc2 Mutant
Diagram indicates the landmarks on upper jaw and lower jaw for the kinematics studies in conditional knockout 4-month-old mice through X-ray Reconstruction of Moving Morphology (XROMM). The position of top is indicated as the temporal and the bottom represents the mandible. Graph indicates the chewing rate of controls (blue) and mutants (red) in kinematics studies carried out in 4-month-old controls and Evc2 mutant mice were quantified. Chewing sequence data can be acquired by measuring the vertical and horizontal displacements.



A- Vertical displacement



B- Anterior Posterior displacement

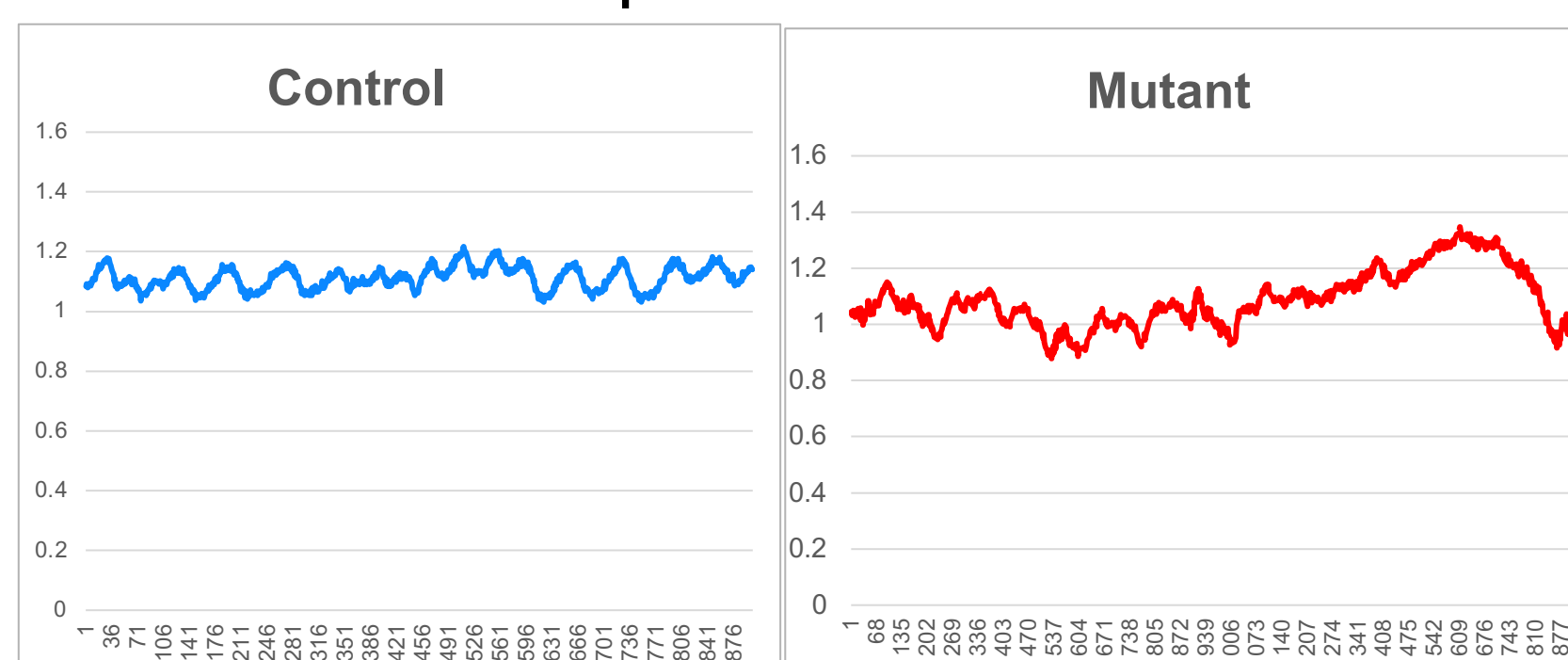


Fig 6. Vertical Displacement and Anterior/ Posterior Movement of Mutant and Control Mice

Track markers for 10~20 cycles of Mandibular and Temporal Data give quantified data on vertical jaw and anterior/posterior jaw movement. Mutant sample with inconsistent and uneven jaw movement, whereas the control model shows consistent jaw movements. The vertical displacement of the control and mutant model. 20 cycles were tracked for the mutant model. On the other hand, 15 cycles were tracked for the control model. The anterior and posterior displacement of control and mutant model became affected more than vertical displacement.

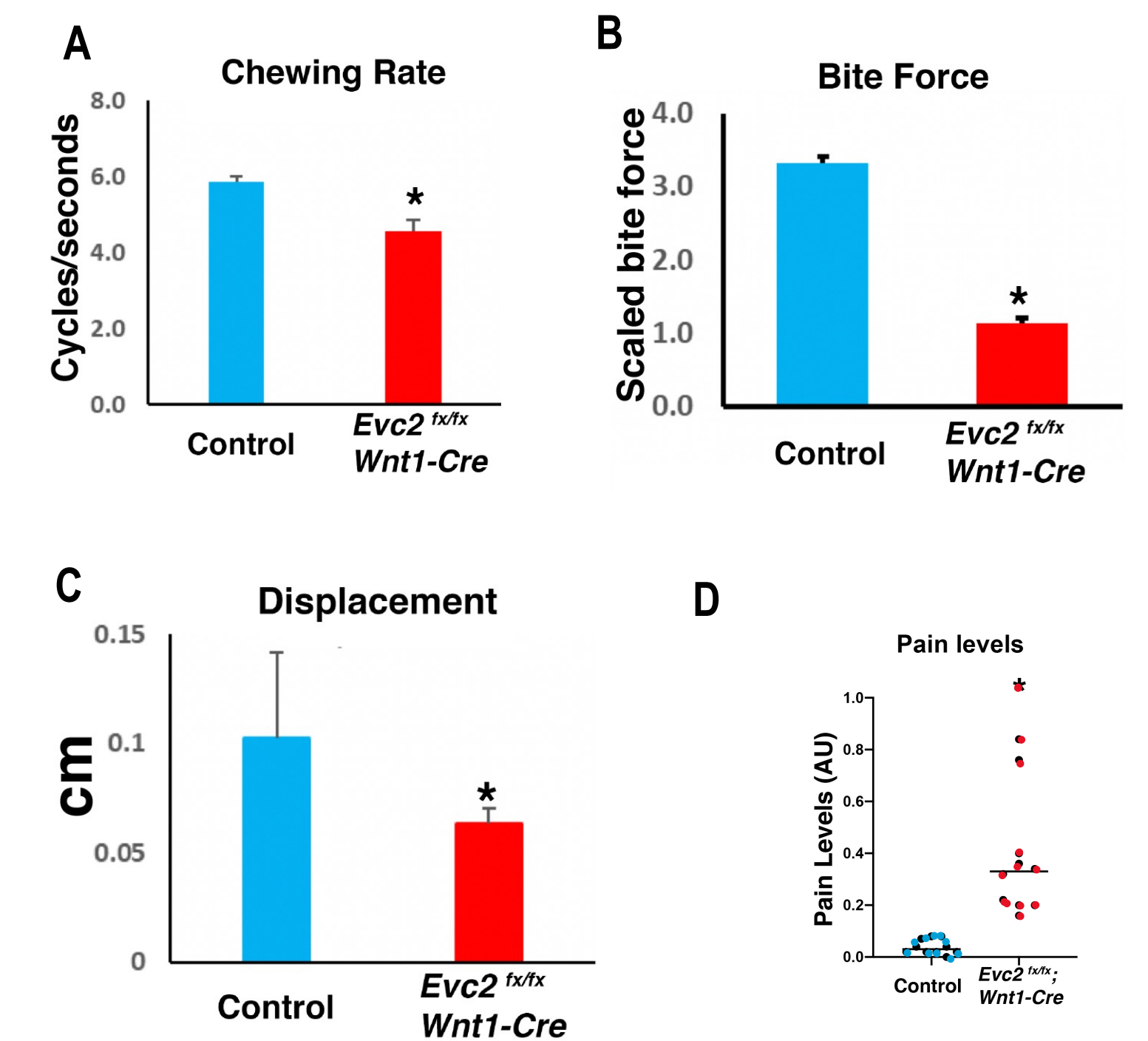


Fig 7. Chart: Displacement, Anti-Posterior Movement, Pain Level and Mastication Rate

Chewing rate (A), Bite force (B), Displacement (C) in kinematics studies carried out in 4-month-old controls and Evc2 mutant mice were quantified. $n=5$, $p<0.01$. D. Pain in controls and Evc2 mutant mice were evaluated through facial grimaces recording for 30 mins and scoring. $N=10$, $p<0.01$

Conclusion

1. Evc2 mutant mice have all characteristics observed in patients with TMD, pain, jaw dysfunction, and structural changes; hence this mutant mouse line is a valuable model to understand the etiology of TMD.
2. Mutant mice are physically incapable of moving the jaw vertically and horizontally in a normal way. Mutant mice in pain, affecting the normal jaw movement we observed in a control group.
3. Control mice have larger, more stable, and more consistent jaw movements vertically and horizontally. Mutant mice chewing pattern is scattered with shorter, smaller, and inconsistent bite range.

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